

HIGH-SWING MOS CURRENT MIRROR WITH ARBITRARILY HIGH OUTPUT RESISTANCE

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A new current mirror circuit scheme is proposed that allows an arbitrary number of transistors to be cascoded creating a current mirror circuit with maximum output voltage swing and very high output resistance.

Introduction: The traditional method for creating a cascode current source or current mirror [1] is shown in Fig. 1. It is composed of two stacked or cascoded chains of transistors, denoted in the Figure as T_1 through T_n and M_1 through M_n . The width-to-length ratio of the gates of transistors T_1 through T_n are not necessarily the same, so we denote their size difference by the real-valued variables x_1^2 through x_{n-1}^2 (square terms are used for mathematical convenience). Transistor T_1 is chosen as the basis for comparison and is denoted as having a width-to-length ratio of W/L . The gates of transistors M_1 through M_n are sized in a similar manner, such that gate of each transistor M_i has the same width and length as the gate of transistor T_i . The current applied to the chain of diode-connected transistors T_1 through T_n creates a set of gate voltages which biases the second chain of transistors M_1 through M_n such that the output current I_{out} is equal to the input current I_{in} .

Linear operation of the cascode current mirror is maintained by ensuring that transistors M_1 through M_n remain in saturation during normal operation. Because transistor M_n is the first transistor in the transistor stack to enter the triode region when the output voltage is reduced, linear operation is guaranteed if the output voltage is greater than the voltage at the gate of M_n minus the threshold voltage V_t . Looking at Fig. 1 we can see that the gate voltage of transistor M_n is equal to

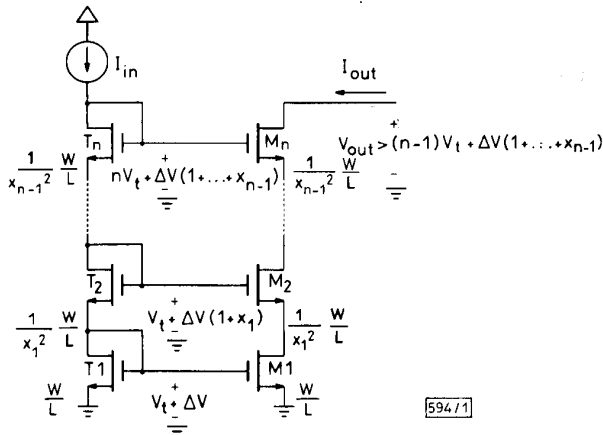


Fig. 1 Generalised version of regular cascode current mirror circuit

the sum of the gate-to-source voltages of transistors T_1 through T_n . Because the gate-to-source voltage of T_1 can be written as $\Delta V + V_t$, with $\Delta V = \sqrt{[I_{in}/K(W/L)]}$, and the gate-to-source voltages of the remaining transistors T_i can be written as $\Delta V \cdot x_{i-1} + V_t$, then the gate voltage of transistor M_n can be seen to be equal to $V_{gsn} = \Delta V(1 + x_1 + \dots + x_{n-1}) + nV_t$. Therefore, for linear operation

$$V_{out} > \Delta V(1 + x_1 + \dots + x_{n-1}) + (n-1)V_t \quad (1)$$

Practically, the number of stacked transistors n is limited to small values to maintain reasonable output voltage swing.

Finally, under small-signal operation, the input and output resistances of the cascode current mirror circuit are given by the following two expressions:

$$r_{in} = \frac{1}{g_m} (1 + x_1^2 + x_2^2 + \dots + x_{n-1}^2) \quad (2)$$

$$r_{out} = r_{oM1} \prod_{i=2}^n g_{mi} r_{oMi} \quad (3)$$

Under conditions of equal-sized transistors, i.e. $x_1 = x_2 = \dots = x_{n-1} = 1$, the input and output resistances can be approximated by $r_{in} = n/g_m$ and $r_{out} = g_m^{n-1} r_{oM1}$, respectively.

High-swing high-output resistance current mirror: Although cascoding transistors increases the output resistance of a current mirror it does so at the cost of reduced linear output voltage range [1]. This stems from the fact that each transistor in the bias circuit is diode-connected, and as a result, is not very efficient at providing the smallest possible drain-source voltage across each transistor to maintain them in saturation. It is easily shown that the drain-to-source voltage of a particular transistor T_i or M_i can be as small as $\Delta V \cdot x_{i-1}$. The circuit shown in Fig. 2 illustrates one way of biasing the transistor chain T_1 through T_n such that the minimum drain-to-source voltages are achieved. This circuit generalises the

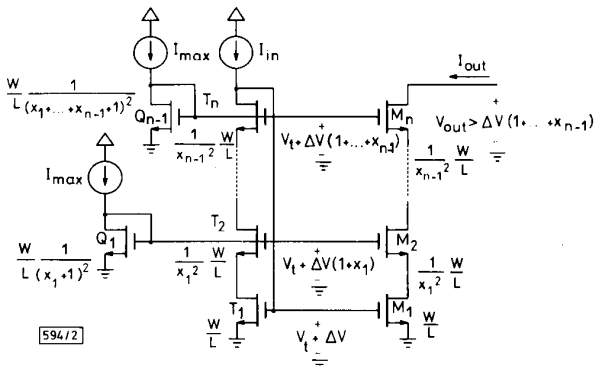


Fig. 2 Generalised version of high-swing cascode current mirror circuit

high-swing single-cascode current mirror described by Babanezhad and Gregorian [2], but first reported by Swanson [3], by allowing more than one level of cascode transistors in the current mirror circuit.

The structure of the high-swing multiple-cascode current mirror circuit shown in Fig. 2 is similar to the cascode current mirror shown previously in Fig. 1 in that it is composed of two chains of stacked transistors. It differs, however, by the way in which the gate voltages of each transistor pair $T_i - M_i$ are formed. The gate voltage of each transistor pair $T_i - M_i$ in the proposed circuit of Fig. 2 (excluding $T_1 - M_1$) is biased independently of each other, unlike that in the cascode current mirror circuit of Fig. 1. The bias voltage for each transistor pair is created using a single diode-connected transistor Q_i fed by a current source I_{max} . As was the case with the original cascode current mirror circuit, the input current I_{in} to the current mirror circuit of Fig. 2 forces the gate-to-source voltage of each transistor T_i to be equal to $\Delta V \cdot x_{i-1} + V_t$, assuming of course that the transistor is in saturation. With independent control over the gate voltage of each transistor, it can ensure that each transistor is biased in its saturation region by forcing the drain-to-source voltage of each transistor to be greater than $\Delta V \cdot x_{i-1}$, i.e. $V_{ds} > V_{gs} - V_t$.

For example, transistor T_1 has a V_{gs} of $\Delta V + V_t$ and therefore to stay in saturation the drain voltage of transistor T_1 must be greater than ΔV . This fixes the value of the gate voltage on T_2 to be greater than $\Delta V \cdot (1 + x_1) + V_t$ because its V_{gs} is equal to $V_t + \Delta V \cdot x_1$. To place this bias voltage on the gate of transistor T_2 the width-to-length ratio of transistor Q_1 is set equal to $[1/(x_1 + 1)^2](W/L)$ (with I_{max} set equal to I_{in}). This process can be repeated up the chain until the gate voltage on transistor T_n is determined. At this point a check is made to ensure that transistor T_n is in saturation, because the drain voltage of T_n is equal to the gate voltage of T_1 . If transistor T_n is not in saturation then the widths of transistors T_1 through T_n will have to be increased. Alternatively, an expres-

sion can be derived that will determine the minimum width-to-length ratios for T_1 through T_n that guarantees that T_n will be in saturation. The constraint on the width-to-length ratio [4] is

$$\frac{W}{L} > \frac{I_{in}(x_1 + \dots + x_{n-1})^2}{KV_t^2} \quad (4)$$

The above described biasing scheme achieves the smallest possible voltage on the gate of transistor M_n . As was the case with the regular cascode current mirror circuit the voltage on the gate of transistor M_n sets the minimum output voltage that maintains linear operation. Following the above described method of calculation, it is found that the gate voltage on M_n is equal to $V_t + \Delta V(1 + x_1 + x_2 + \dots + x_{n-1})$ and therefore the output voltage range is

$$V_{out} > \Delta V(1 + x_1 + x_2 + \dots + x_{n-1}) \quad (5)$$

Thus the output voltage range of the high-swing cascode current mirror circuit is larger than the cascode current mirror circuit of Fig. 1 by a term of $(n-1)V_t$. Note that previously [2, 3] this result was known only for $n=2$.

The small-signal operation of the high-swing cascode current mirror circuit is similar to that of the regular cascode current mirror circuit, with the exception that the input resistance is found to be about n times smaller.

Simulation and experimental results: To verify the predicted increases in the output voltage swing, a comparison between the large-signal DC transfer characteristic I_{out} against V_{out} for a double-cascode current mirror and high-swing double-cascode current mirror was made. The input bias current I_{in} was chosen to be $10 \mu A$ in each case. The transistors were all made to have equal dimensions of $W = 9 \mu m$ and $L = 1.2 \mu m$. The transistor model parameters were based on transistors fabricated using the $1.2 \mu m$ CMOS process of Northern Telecom Ltd. The zero-bias threshold voltage V_t is $0.8 V$. The SPICE results are shown in Fig. 3. The solid line indicates the results for the regular double-cascode current mirror circuit (Fig. 1, $n=3$) and the dotted line indicates the results for the high-swing double-cascode current mirror circuit (Fig. 2, $n=3$). On inspection of the SPICE results, we find that the minimum output voltage for the double-cascode current mirror circuit that maintains linear operation is about $2.0 V$ and that found for the high-swing double-cascode circuit is $0.35 V$. The difference between these two minima is $1.65 V$ which is only slightly higher than the expected value of $2V_t$. At an output voltage of $2.5 V$, a SPICE small-signal analysis indicates that the output resistance of the double-cascode mirror

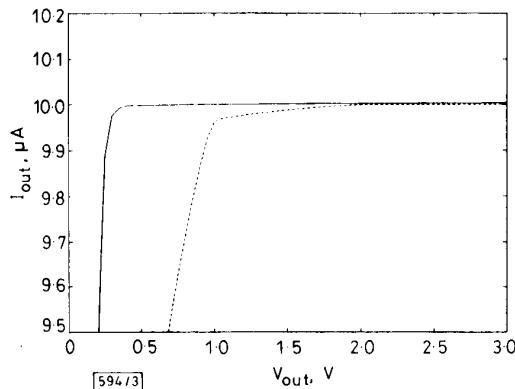


Fig. 3 Simulation results for regular and high-swing double-cascode current mirror circuits

$I_{in} = 10 \mu A$
 regular cascode
 — high-swing cascode

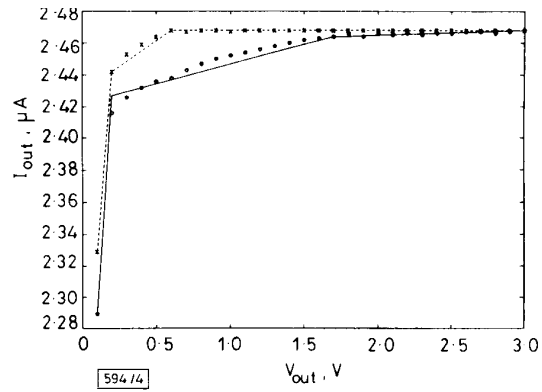


Fig. 4 Experimental results for regular and high-swing double-cascode current mirror circuits

$I_{in} = 2.45 \mu A$
 ○ regular cascode
 × high-swing cascode

is $10.3 G\Omega$. Likewise, the output resistance for the high-swing double-cascode mirror is $30.8 G\Omega$. Comparing this to the resistance values obtained for the single-cascode circuit and the high-swing single-cascode circuit (35 and $25 M\Omega$, respectively), we see that the output resistance of either double-cascode current mirror circuit is about three orders of magnitude larger.

Discrete prototypes of variations of the above current mirror circuits were also constructed. Individual MOS transistors were derived from the 4007 CMOS logic inverter IC supplied by RCA. Owing to transistor sizing restrictions, the bias voltages for the high-swing double-cascode current mirror were derived from various resistor divider circuits. It was also found necessary to select bias currents that were slightly different from the simulations due to the size of the devices used. The measured results are shown collectively in Fig. 4. Close correlation is seen to exist between the simulation results and those obtained by measurement. Direct comparisons cannot be made as the models for the transistors on the 4007 CMOS logic inverter circuit were not obtained.

Conclusions: This Letter generalises the scheme used to create a high-swing single-cascode current mirror circuit so that it may be applied to the case of a multiple cascode circuit. Using this new technique it is possible to create current sources and current mirrors with very high output resistance together with maximum output voltage swing.

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